

Mini Assignment: String in Python

Assignment 1: Password Security Analyzer

Problem Statement

A cybersecurity company wants to analyze user passwords before allowing account creation.

The system should accept at least **15 passwords** and generate a security report.

Requirements

For each password:

1. Count uppercase letters.
2. Count lowercase letters.
3. Count digits.
4. Count special characters.
5. Check minimum length (8 characters).
6. Check if spaces exist.
7. Determine password strength:
 - Strong
 - Medium
 - Weak
8. Display repeated characters.
9. Count vowels and consonants.
10. Identify the most frequently occurring character.

Challenge

Generate a report showing:

Total Passwords Analyzed

Strong Passwords

Medium Passwords

Weak Passwords

Assignment 2: Email Validation & Domain Analytics System

Problem Statement

An organization has collected **20 email addresses** from users.

Create a program to analyze these email addresses.

Requirements

For each email:

1. Extract username.
2. Extract domain.

3. Extract extension.
4. Count digits in username.
5. Count special characters.
6. Check if email is valid:
 - Exactly one '@'
 - Contains '.'
 - No spaces
7. Display invalid emails.
8. Count emails belonging to each domain.

Sample Input

rahul123@gmail.com
priya@outlook.com
anuj@company.in

Challenge

Generate a domain report:

gmail.com -> 8 users
outlook.com -> 5 users
yahoo.com -> 3 users
company.in -> 4 users

Assignment 3: Chat Message Analytics Dashboard

Problem Statement

A messaging application wants to analyze chat messages.

Store at least **20 chat messages** in a list.

Requirements

For each message:

1. Count total words.
2. Count total characters.
3. Count vowels and consonants.
4. Find longest word.
5. Find shortest word.
6. Count occurrence of each word.
7. Display repeated words.
8. Display words starting with vowels.
9. Display words longer than 5 characters.
10. Create a dictionary containing word frequencies.

Challenge

Generate a report showing:

Most Frequently Used Word

Longest Message

Shortest Message

Average Words Per Message

Assignment 4: Vehicle Registration Verification System

Problem Statement

A transport department wants to verify vehicle registration numbers.

Store at least **20 vehicle numbers**.

Example

MH12AB4589

DL05XY9988

KA03PQ1234

Requirements

For each registration number:

1. Extract state code.
2. Extract district code.
3. Extract series.
4. Extract vehicle number.
5. Count letters and digits.
6. Validate format:
 - First 2 characters = Alphabets
 - Next 2 characters = Digits
 - Next 2 characters = Alphabets
 - Last 4 characters = Digits
7. Display invalid registrations.
8. Count vehicles state-wise.

Challenge

Generate a state-wise report:

MH -> 6 Vehicles

DL -> 4 Vehicles

KA -> 5 Vehicles

UP -> 5 Vehicles

Assignment 5: News Article Text Analyzer

Problem Statement

A news agency wants to analyze the content of an article.

Use a paragraph containing **at least 300 words**.

Requirements

1. Count total characters.
2. Count total words.
3. Count total sentences.
4. Count vowels and consonants.
5. Find longest word.
6. Find shortest word.
7. Find the most frequent word.
8. Create a dictionary of word frequencies.
9. Display words appearing only once.
10. Display words appearing more than 5 times.
11. Count words starting with each alphabet.
12. Display all unique words.

Challenge

Generate a complete text summary:

Total Words

Total Sentences

Average Word Length

Most Frequent Word

Vocabulary Size

Assignment 6: Student Resume Analyzer

Problem Statement

A student enters a resume as plain text (Name, Skills, Education, Projects, Achievements).

The system should:

1. Count total words.
2. Count total characters.
3. Extract email IDs.
4. Extract phone numbers.
5. Count skills mentioned.
6. Find repeated keywords.
7. Identify the most frequently used word.
8. Generate a skill frequency report.

9. Detect duplicate skills.
10. Create a summary dashboard.

Expected Output

Resume Analysis Report

Total Words: 420

Total Characters: 2650

Email Found: 1

Phone Numbers Found: 1

Most Frequent Skill: Python

Top 5 Keywords:

Python

SQL

React

Java

Communication